Week 8 : 19 Oct 2026 to 23 Oct 2026

Subject/الموضوع: MYP Biology **Unit Title/عنوان الوحدة:** Unit 6 – The Human Body (GRASPS: Heart Rate Investigation) **Grade/الصف:** 10

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: 1 الحصة **GRASPS Launch: Heart-Rate Control** تنظيم معدل ضربات القلب	• I can explain how the circulatory system and feedback mechanisms regulate heart rate to maintain homeostasis.	7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 6.1 Guide goal-setting with prompts, examples, and checklists.	All resources are shared on teams/toddle
Lesson 2: 2 الحصة **Designing the Investigation** تصميم الاستقصاء العلمي	• I can design a safe, valid investigation to test how exercise affects heart rate and recovery.	6.2 Support planning with templates, strategy prompts, and modeled thinking. 5.3 Build fluency with graduated scaffolds and models. 8.3 Foster collaboration and community with structured group roles and supports. **Formative:** Teacher sign-off on hypothesis, variables & method (Crit. B). **Differentiation:** Tiered planning templates; SEN: pre-filled method skeleton.	All resources are shared on teams/toddle
Homework الواجبات	• Report Section 1: how heart rate is regulated (150–200 words) – Toddle. • Report Section 2: question, hypothesis, variables & method – Toddle.	6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides. 5.2 Use multiple tools for composition (spellcheckers, manipulatives, templates). 8.4 Provide mastery-oriented feedback that emphasizes growth and improvement.	All resources are shared on teams/toddle
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	• 2-week GRASPS summative task (4 sessions): Week 8 = Criteria A & B; Week 9 = Criteria C & D. • Light exercise next week (step-ups / jumping jacks) – please inform the teacher of any health concerns.		

Week 8 الأسبوع8

Technology Integration دمج التكنولوجيا	• Toddle: task sheet, rubrics, draft submission and feedback. • McGraw-Hill Inspire (ConnectED): Module 24 e-book, LearnSmart, Circulatory System Virtual Investigation.
Cross-curricular links الروابط بين المواد الدراسية	• **Mathematics:** converting 15-second pulse counts to beats per minute (×4). • **Physical Education:** exercise intensity, warm-up and recovery. • **English:** scientific writing – research question and IF–THEN–BECAUSE hypothesis.
External Assessment Integration دمج التقييمات الخارجية	• **MAP Growth Science:** Life Science – structure, function and information processing; scientific inquiry (variables, hypotheses).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• **Thinking (critical):** analyse feedback-loop diagrams to predict body responses. • **Research (inquiry):** identify and control variables; plan data collection. • **Social (collaboration):** team roles – lead, timer, recorder, safety officer.
Standards المعايير	• HS-LS1-3: Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis. • HS-LS1-2: Develop and use a model to illustrate the hierarchical organization of interacting systems. • MYP Criterion A (i, ii, iii) and Criterion B (i, ii, iii, iv).
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	• **Factual:** What is homeostasis, and how does a feedback mechanism work? • **Debatable:** How should a society balance personal lifestyle choices against the public-health burden of preventable disease?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Lesson 1 – GRASPS Launch: Heart-Rate Control** • **MUST:** I can describe the path of blood through the heart and body and state the function of the heart, arteries, veins and capillaries. • **SHOULD:** I can explain, using a negative-feedback loop (stimulus → receptor → control centre → effector → response), how heart rate rises during exercise and returns to resting rate during recovery. • **COULD:** I can analyse how the circulatory, respiratory and nervous systems interact to keep blood O_2/CO_2 balanced, and predict how fitness level changes recovery time. **Lesson 2 – Designing the Investigation** • **MUST:** I can write a testable research question and identify the independent, dependent and at least two controlled variables. • **SHOULD:** I can write an IF–THEN–BECAUSE hypothesis using circulatory-system and homeostasis reasoning, and describe a safe, step-by-step method to measure pulse at rest, after exercise and during recovery. • **COULD:** I can justify how each controlled variable will be managed, plan repeat trials to improve reliability, and explain how sufficient, relevant data will be collected.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Compassion التعاطف Arabic Language اللغة العربية المجال Area: Citizenship – Belonging | Values – Compassion | Culture – Arabic Language التطبيق Implementation: **Belonging:** researching heart health for a UAE clinic and community. **Compassion:** how heart disease affects UAE families. **Arabic Language:** bilingual key-term glossary (قلب، نبض، الاتزان الداخلي).